

1 Solution — GIAM 1.7

Problem 5

1.1 Problem

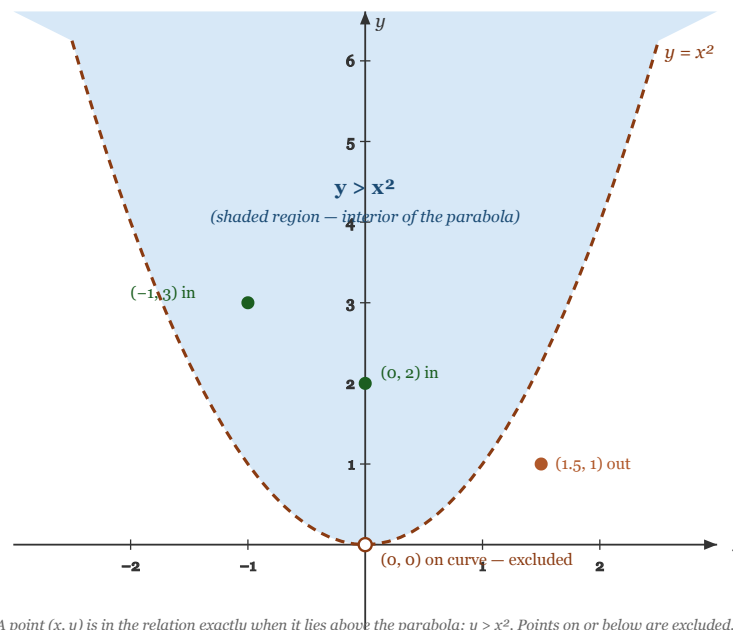
Sketch a graph of the relation

$$\{ (x, y) \mid x, y \in \mathbb{R} \text{ and } y > x^2 \}.$$

1.2 The Sketch

The relation is the set of all points lying **strictly above** the parabola $y = x^2$ — the parabola's *interior*, with the curve itself excluded:

The set of points strictly above the parabola $y = x^2$. The curve itself is dashed — it is not included.



A point (x, y) is in the relation exactly when it lies above the parabola: $y > x^2$. Points on or below are excluded.

Figure 1.1: A Cartesian graph. The parabola $y = x^2$ opens upward with vertex at the origin, drawn as a dashed brown curve because the boundary is not part of the relation. The region above and inside the parabola is shaded light blue and labelled $y > x^2$. Sample points: $(0, 2)$ and $(-1, 3)$ are inside the shaded region (they satisfy $y > x^2$); $(0, 0)$ sits on the curve and is drawn as an

open circle to show it is excluded (0 is not > 0); (1.5,1) is below the curve and outside the region (1 is not > 2.25).

1.3 How to Read the Sketch

- **The boundary** $y = x^2$ is drawn **dashed**. Because the inequality is *strict* ($>$, not $\geq$), points *on* the parabola are **not** in the relation.
- **The shaded region** is everything *above* the parabola. For a given x , the condition $y > x^2$ says " y is higher than the parabola's height at that x ," so we take all points above the curve.
- **Test a point.** The origin's neighbour $(0, 2)$ gives $2 > 0^2 = 0$ ✓ (inside). The point $(1.5, 1)$ gives $1 > 1.5^2 = 2.25$, which is false ✗ (outside). The vertex $(0, 0)$ gives $0 > 0$, false — on the curve, hence excluded.

1.4 Remark — Dashed vs. Solid: The Boundary Convention

The single most important sketching decision here is dashing the parabola. A relation defined by a **strict** inequality ($<$ or $>$) excludes its boundary, drawn as a *dashed/open* curve; a **non-strict** inequality ($\leq$ or $\geq$) includes it, drawn *solid*. Had the problem said $y \geq x^2$, the same region would be shaded but the parabola would be solid and *part of* the set. This mirrors exactly the open-versus-closed distinction from the number-line intervals of Chapter 1: " $>$ " is an open boundary, just as (a, ∞) uses a hollow endpoint.

1.5 Remark — This Relation Is Emphatically Not a Function

Unlike Problem 2's $f(x) = x^2 \bmod 7$, this relation is **not a function**. For almost every x , infinitely many y satisfy $y > x^2$ — e.g. at $x = 0$ every positive y qualifies. So a single input is paired with a whole vertical ray of outputs, and the "vertical line test" fails spectacularly: a vertical line $x = c$ meets the region in an entire half-line, not a single point. This is a good reminder that "relation" is far more general than "function"; functions are the special relations where each x has *exactly one* partner y , whereas here each x has infinitely many.

1.6 Remark — Domain and Range of the Region

Reading off the picture in the language of Problem 2: the **domain** (the x -values that appear) is *all* of $\mathbb{R}$ — every vertical line eventually rises above the parabola, so some point sits over each x . The **range** (the y -values that appear) is $(0, \infty)$: every positive y is achieved (take $x = 0$), but no $y \leq 0$ can satisfy $y > x^2 \geq 0$. So although the boundary point $(0, 0)$ is excluded, the region still reaches arbitrarily *close* to it and to the whole nonnegative x -axis — the set is **open**, containing none of its edge.

1.7 Remark — Slicing the Region

A clean way to *describe* the same set without a picture is by horizontal and vertical slices. **Vertically**, above each fixed x the allowed y form the open ray (x^2, ∞) . **Horizontally**, at a fixed height y the condition $y > x^2$ rearranges to $x^2 < y$, i.e. $-\sqrt{y} < x < \sqrt{y}$ (only possible when $y > 0$) — an open interval that widens as y grows, which is exactly why the shaded region flares outward going up. Both slice-descriptions encode the identical region; the sketch simply shows all the slices at once.